114-2 電工實驗（通信專題）

Software-Defined Radio (SDR)

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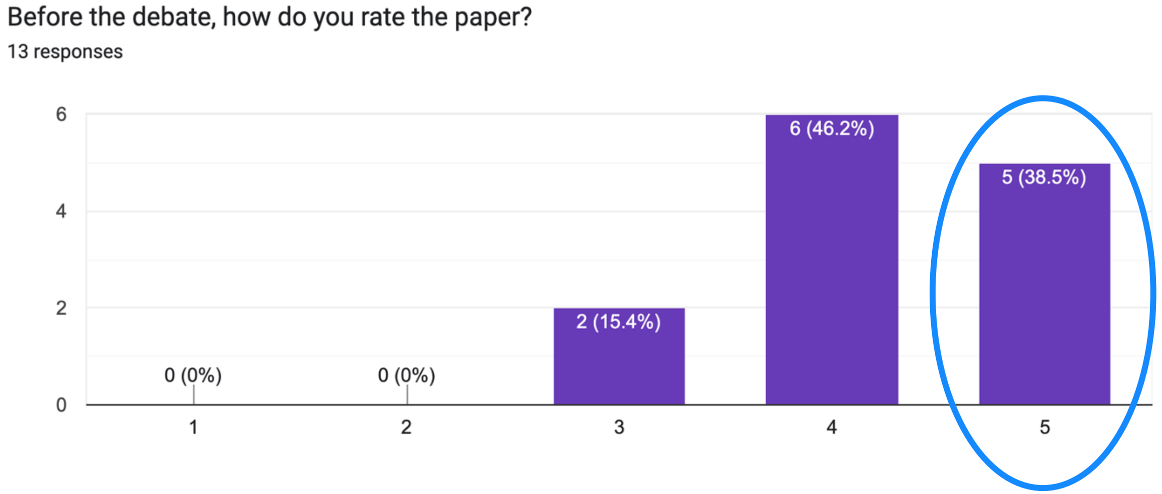
Paper 2 Debate Review

Feedback from the audience

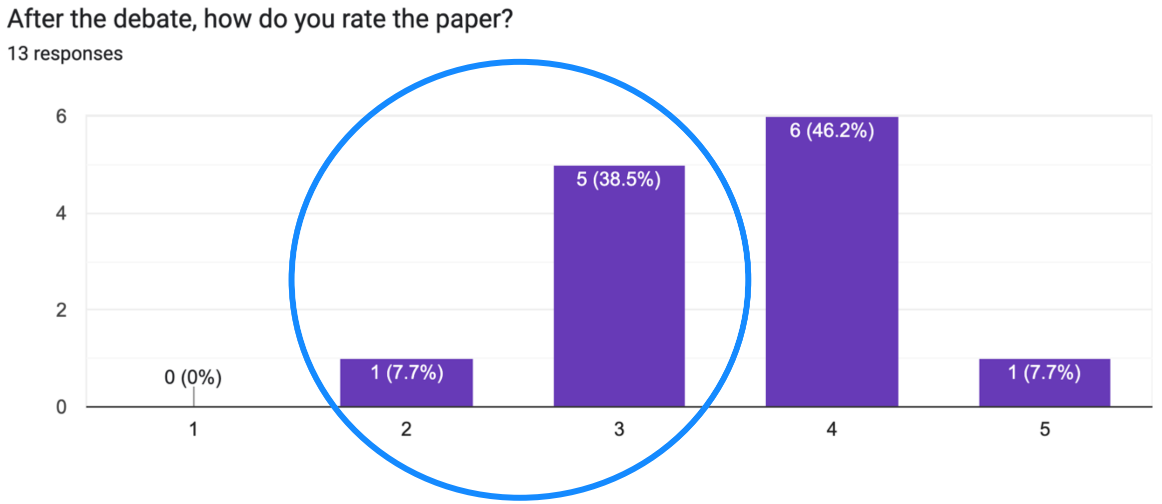
- 正方：有自己做解釋phase drift方法的圖 不錯 整體講解清楚；對論文定位清楚→軟體solution而非硬體
- 反方：論點清楚、對debug mode future work 可能性的反駁也很合理
- 正方講的很完整 反方的論點也有攻擊性

How the paper is rated among the class



Before the debate

Average = 4.23



After the debate

Average = 3.54

沒有經驗容易被唬住,

多經歷幾次就會變得很精明！

Software-Defined Radio

What is it?

How is it different from a pure hardware approach?

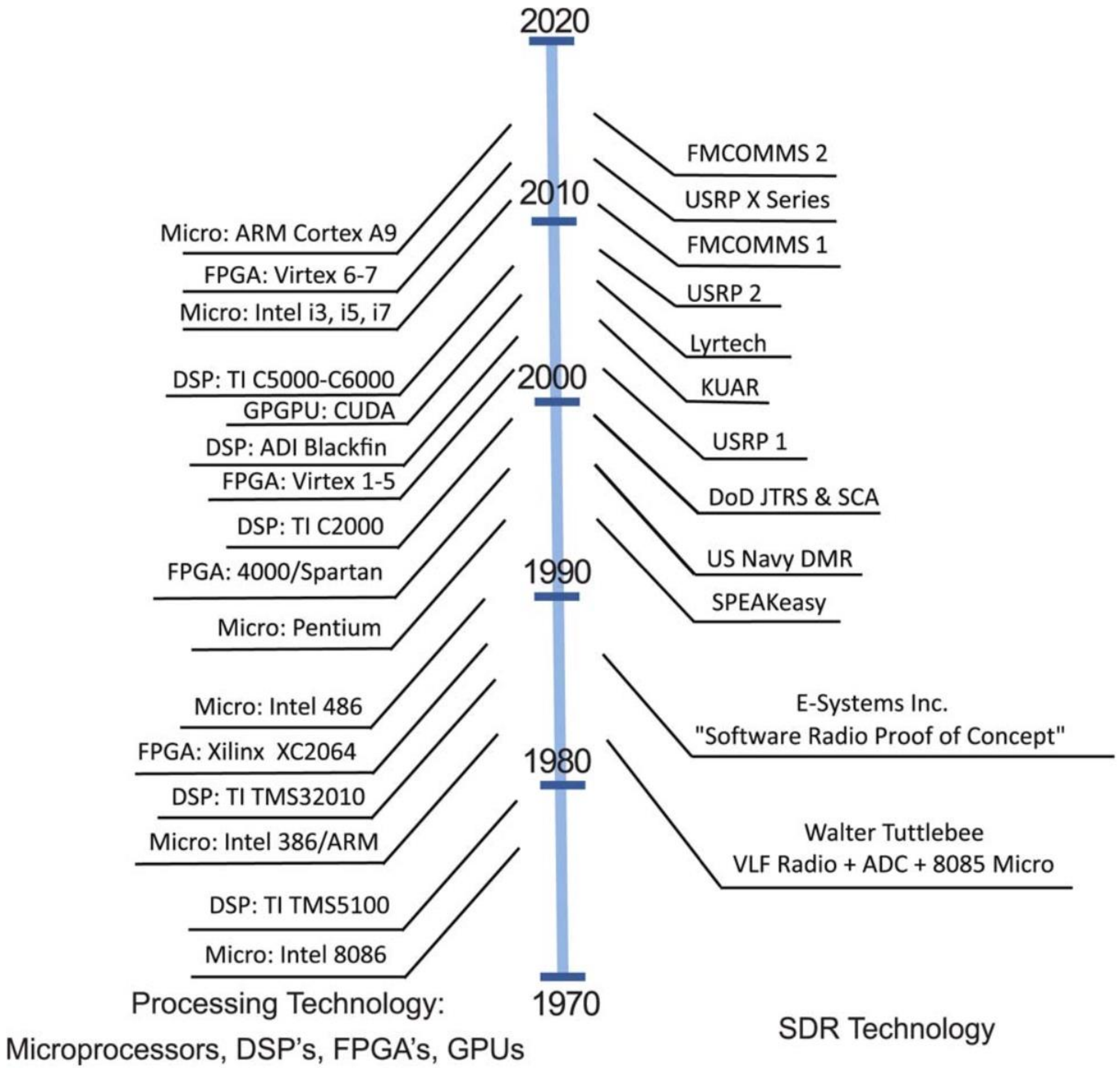
What can it do?

History

- 1950s: wireless systems operated exclusively in the **analog** domain, where communications functions such as modulation and filtering were performed using analog circuits and components
- Rapid evolution of **digital** technology - especially analog-to-digital and digital-to-analog converters
  ➔ perform these same baseband communication functions partially or entirely within the digital domain
  - reducing cost
  - enabling mass production of these transceivers
  - providing greater flexibility and system functionality

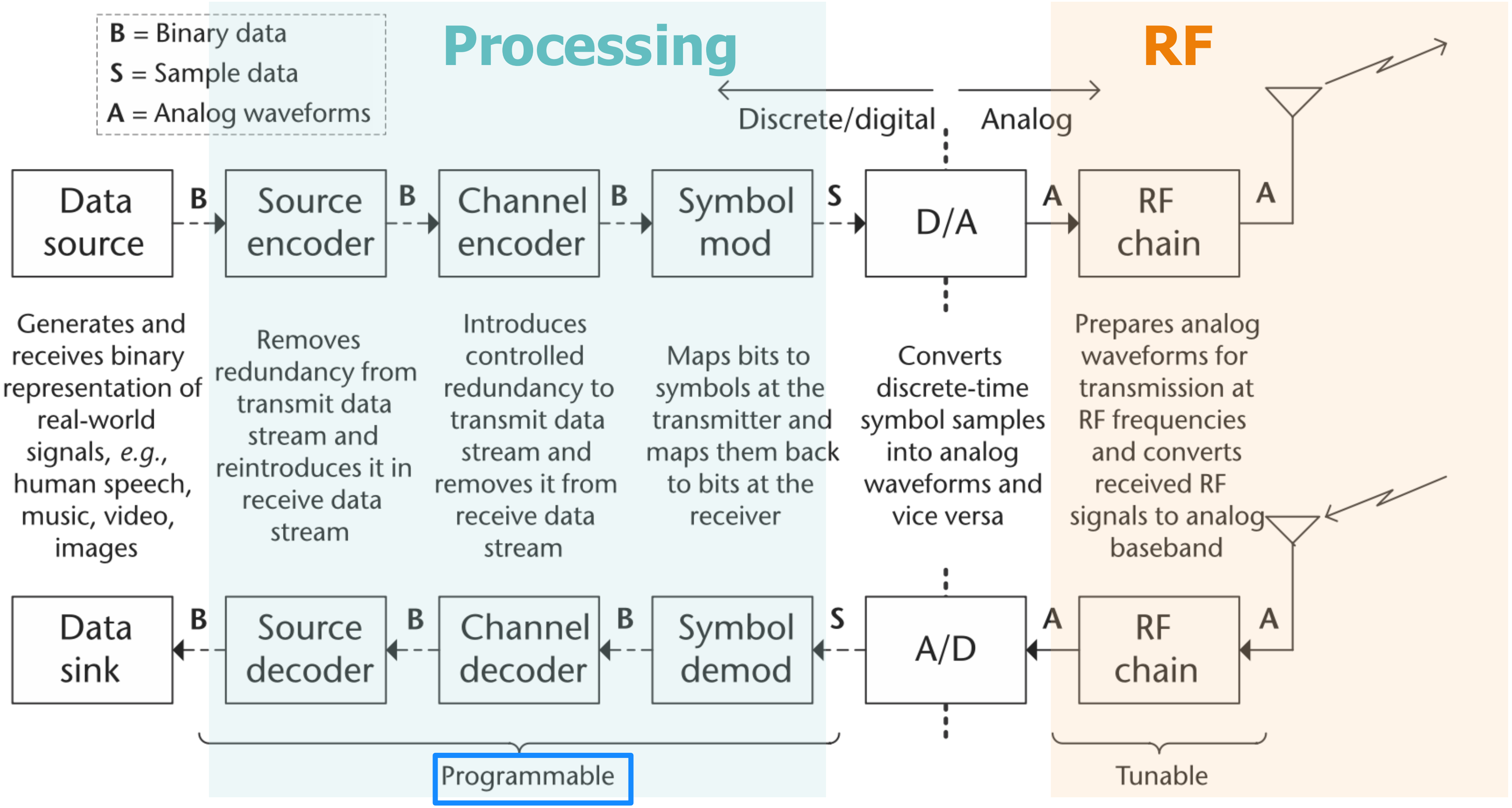
History

- First: **application-specific integrated circuits (ASICs)**.
  - nonprogrammable, static designs
  - cellular telephones, Wi-Fi modems
- 1970s: wireless transceivers that possessed programmable (or software-defined) attributes.
- 1980s: digital baseband radios with programmable features were starting to be prototyped
- 1990s: the first large-scale SDR platforms, SpeakEASY 1 and SpeakEASY 2.
  - Many computing technologies were used: multiple **DSP** platforms and **field programmable gate array** (**FPGA**) technology


2020
2010
2000
1990
1980
1970
Micro: ARM Cortex A9
FPGA: Virtex 6-7
Micro: Intel i3, i5, i7
DSP: TI C5000-C6000
GPGPU: CUDA
DSP: ADI Blackfin
FPGA: Virtex 1-5
DSP: TI C2000
FPGA: 4000/Spartan
Micro: Pentium
Micro: Intel 486
FPGA: Xilinx XC2064
DSP: TI TMS32010
Micro: Intel 386/ARM
DSP: TI TMS5100
Micro: Intel 8086
FMCOMMS 2
USRP X Series
FMCOMMS 1
USRP 2
Lyrtech
KUAR
USRP 1
DoD JTRS & SCA
US Navy DMR
SPEAKeasy
E-Systems Inc.
"Software Radio Proof of Concept"
Walter Tuttlebee
VLF Radio + ADC + 8085 Micro
Processing Technology:
Microprocessors, DSP's, FPGA's, GPUs
SDR Technology

Software-Defined Radio (SDR)

Radio in which some or all of the physical layer functions are software defined.



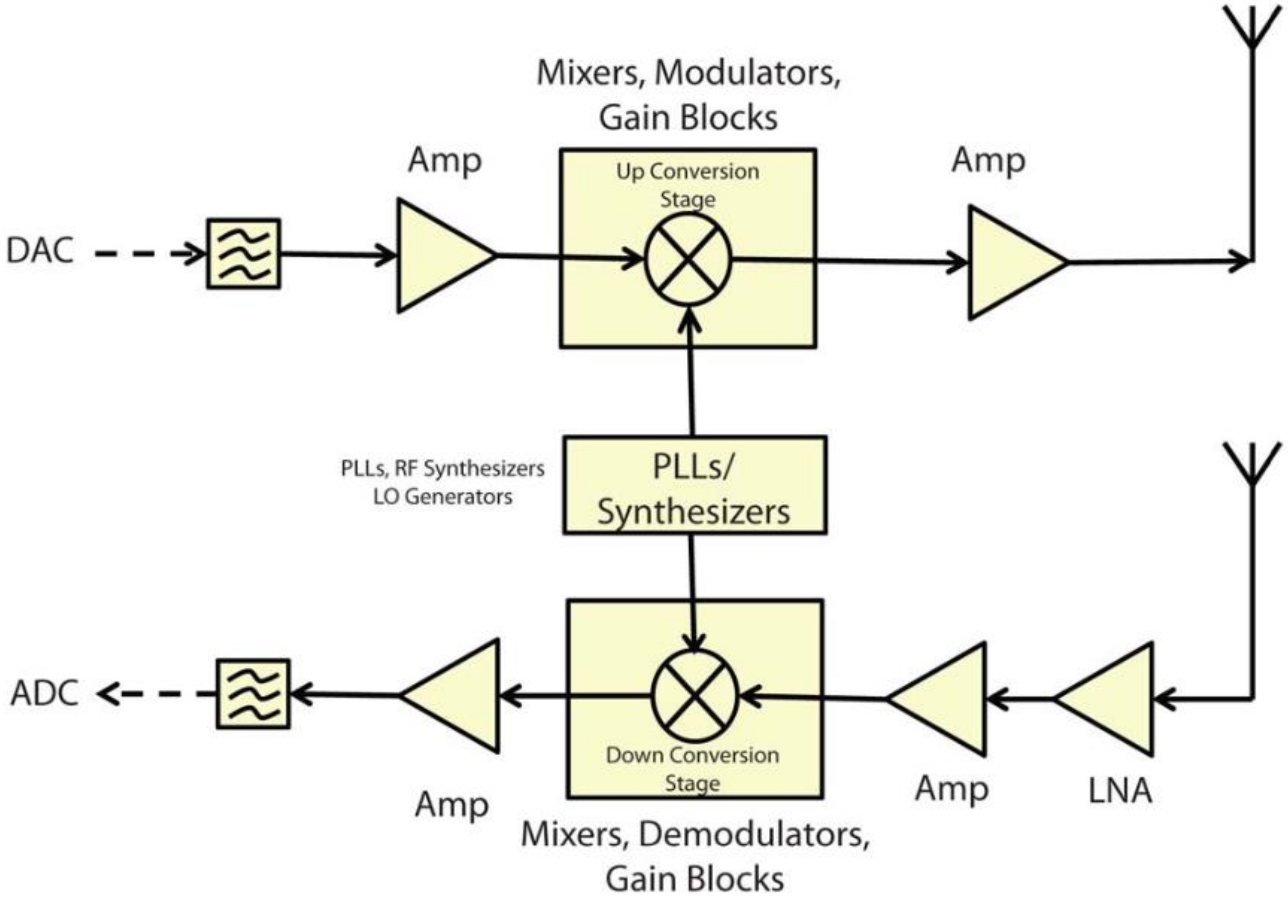
can be realized in either programmable logic or software.

Not limited to these blocks

RF Architectures

Up-conversion:

$$y(t) = Re\{x(t)e^{j2\pi f_c t}\}$$



Digital PLL Frequency Synthesizers: what they are, how they work by ElectronicsNotes

https://youtu.be/5K7Pvc5fxZI?si=MzKL6gLE3QMq7Q-d

Software Architectures

General-Purpose Microprocessors

+ high level of flexibility with respect to reconfigurability.
- not specialized for mathematical computations, they can be very power inefficient.

Digital signal processors (DSPs)

+ Perform specialized mathematical computations. Implement new modules with relative ease, relatively power-efficient
- not well-suited for computationally intensive processes and can quickly lose speed

Field programmable gate arrays (FPGAs)

+ Computationally powerful, relatively power efficient
- Inflexible, difficult to implement new modules

Graphics processing units (GPUs)

+ Computationally powerful
- Difficult to use, especially when trying to implement new modules

Software Environment

Two-step development process

What we do in this class

1. Develop, tune, and optimize the modulation and demodulation algorithms for a specific sample rate, bandwidth, and environment. Normally done on a host PC.
2. Take the above algorithm, which may be implemented in a high-level language in floating point, and code it in a production-worthy environment, making production trade-offs of a product's size, weight, power and cost (SWaP-C) in mind.

Common Options

- MATLAB
  - Cross-platform (Windows, Linux, MAC) offering support for many of the popular commercial radio front-ends.
  - Simulink: real-world system simulation and automatic code generation for hardware and software implementation.
- GNU Radio
  - Open source: C++, Python

Supported Hardware – Software-Defined Radio

R2024b

Support for third-party software-defined radio hardware, such as RTL-SDR, ADALM-PLUTO, and USRP™ radios

Get Support Package Now

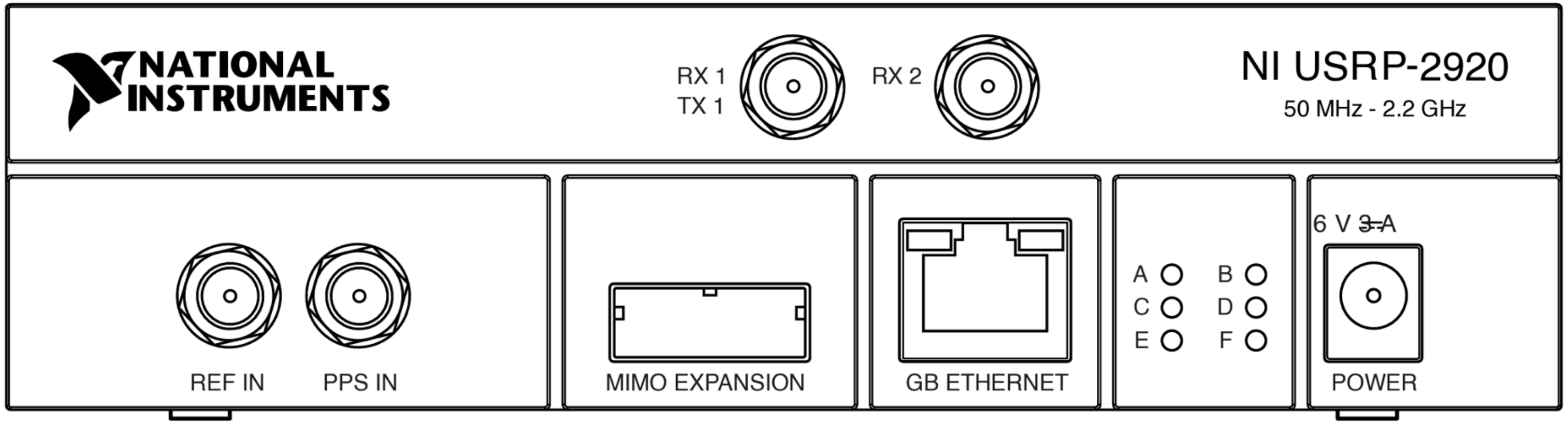
Communications Toolbox™ supports this hardware.

Support Package	Vendor	Earliest Release Available	Latest Release Available
ADALM-Pluto Radio	Analog Devices®	R2017a	Current
RTL-SDR Radio	NooElec®	R2013b	Current
USRP Embedded Series Radio	Ettus Research™	R2016b	Current
USRP Radio	Ettus Research	R2011b	Current
Xilinx® Zynq®-Based Radio	Xilinx	R2014b	R2023b **Note:** Starting in R2024a, see Radio Applications (SoC Blockset)

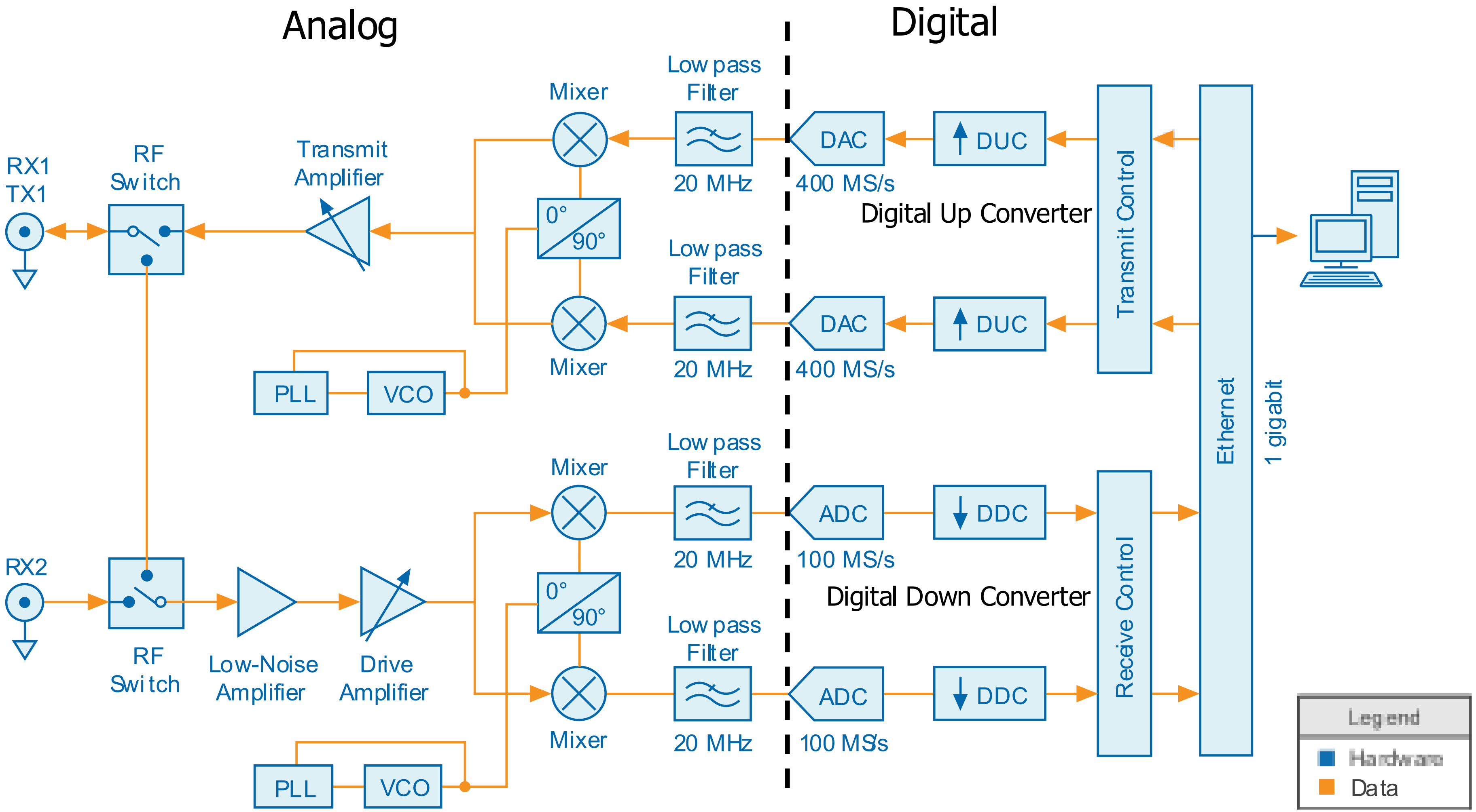
https://www.mathworks.com/help/comm/supported-hardware-software-defined-radio.html

Universal Software Radio Peripheral

USRP-2920

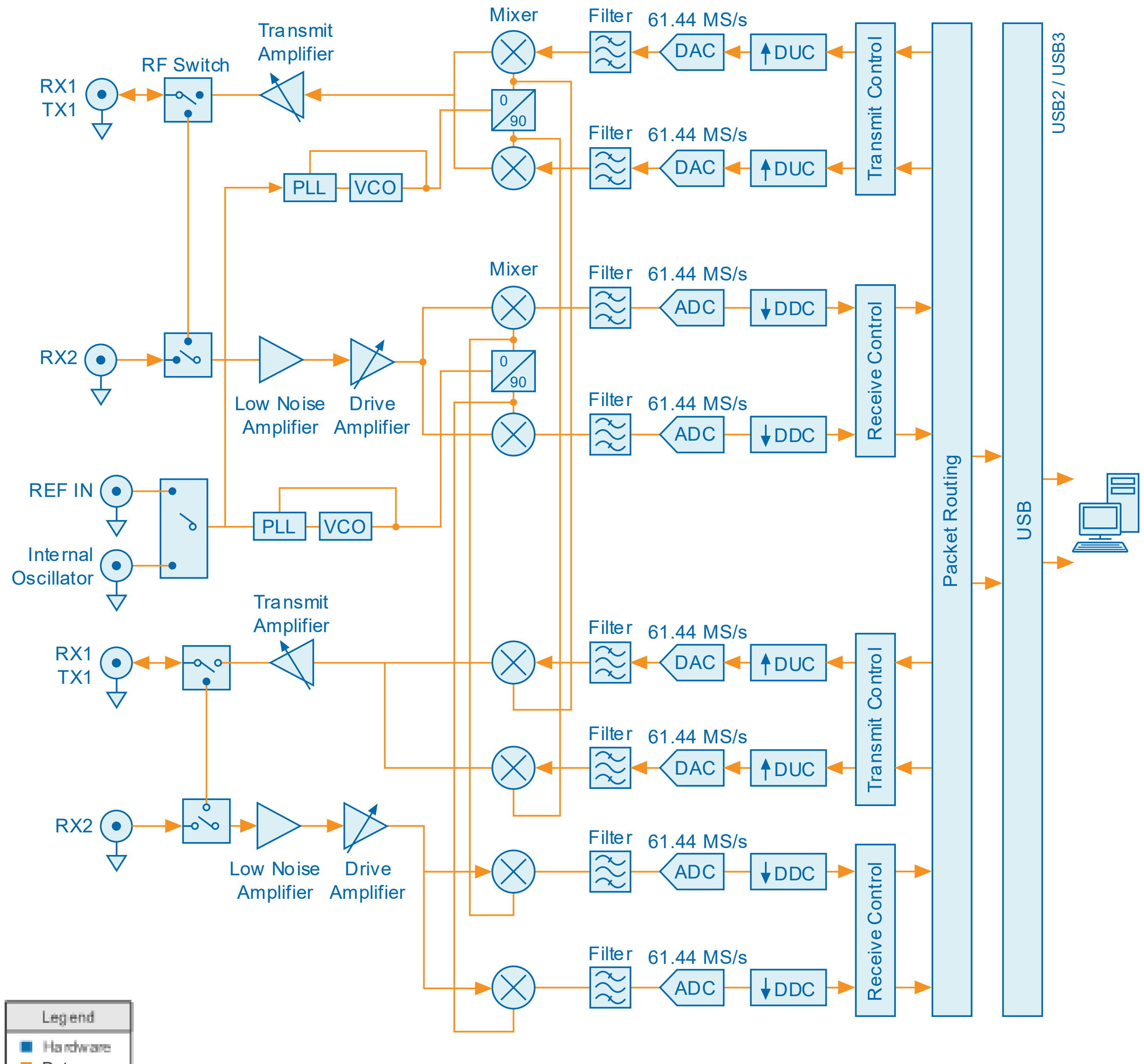


Block Diagram of USRP-2920



Can we use both antennas to receive simultaneously? No!

Block Diagram of USRP-2901



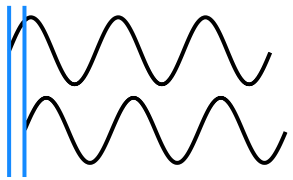
Note: The two RX channels share a single oscillator, and the two TX channels share a single oscillator.

Can we use all 4 antennas simultaneously?

How many TX, RX?

Do the two TX channels share the same oscillator? Why is it important?

If two TX use different oscillators that exhibit a phase difference ϕ (ϕ can vary each time the device is turned on)



Up-conversion:

$$y_1(t) = Re\{x(t)e^{j2\pi f_c t}\}$$

$$y_2(t) = Re\{x(t)e^{j2\pi f_c t+\phi}\}$$

$$= Re\{x(t)\boxed{e^{\phi}} \cdot e^{j2\pi f_c t}\}$$

Phase

Reference - SDR

- Travis Collins, Robin Getz, Alexander Wyglinski, Di Pu. *Software-Defined Radio for Engineers*. Artech, 2018. [NTU Library Link]

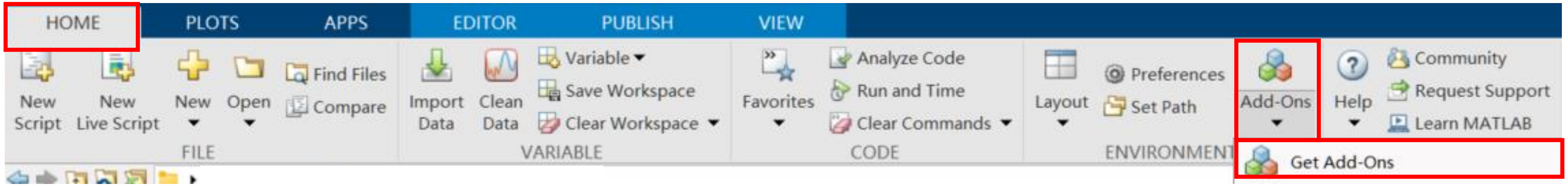
- Raquel G. Machado and Alexander M. Wyglinski. "Software-defined radio: Bridging the analog–digital divide." *Proceedings of the IEEE* 103, no. 3 (2015): 409-423.

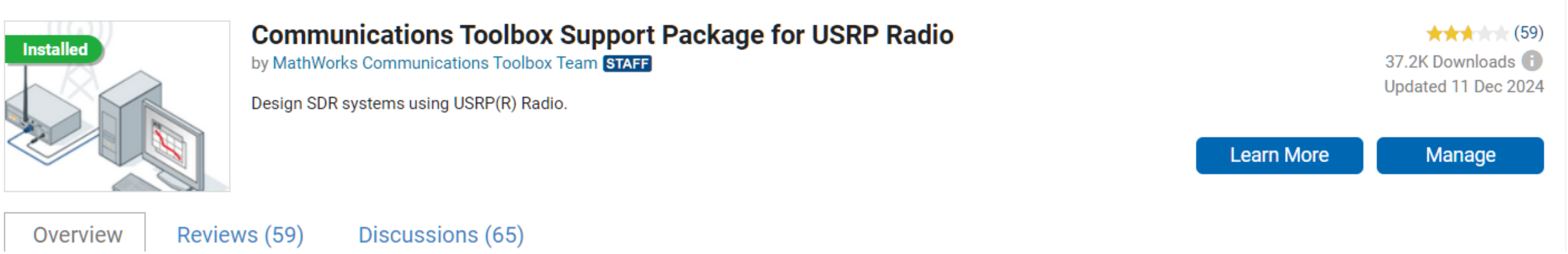
USRP 2920 Example Code

By TA 邑恒、TA 奕昕

USRP Preparation - Software

- Software : Matlab Communications Toolbox Support Package for USRP Radio





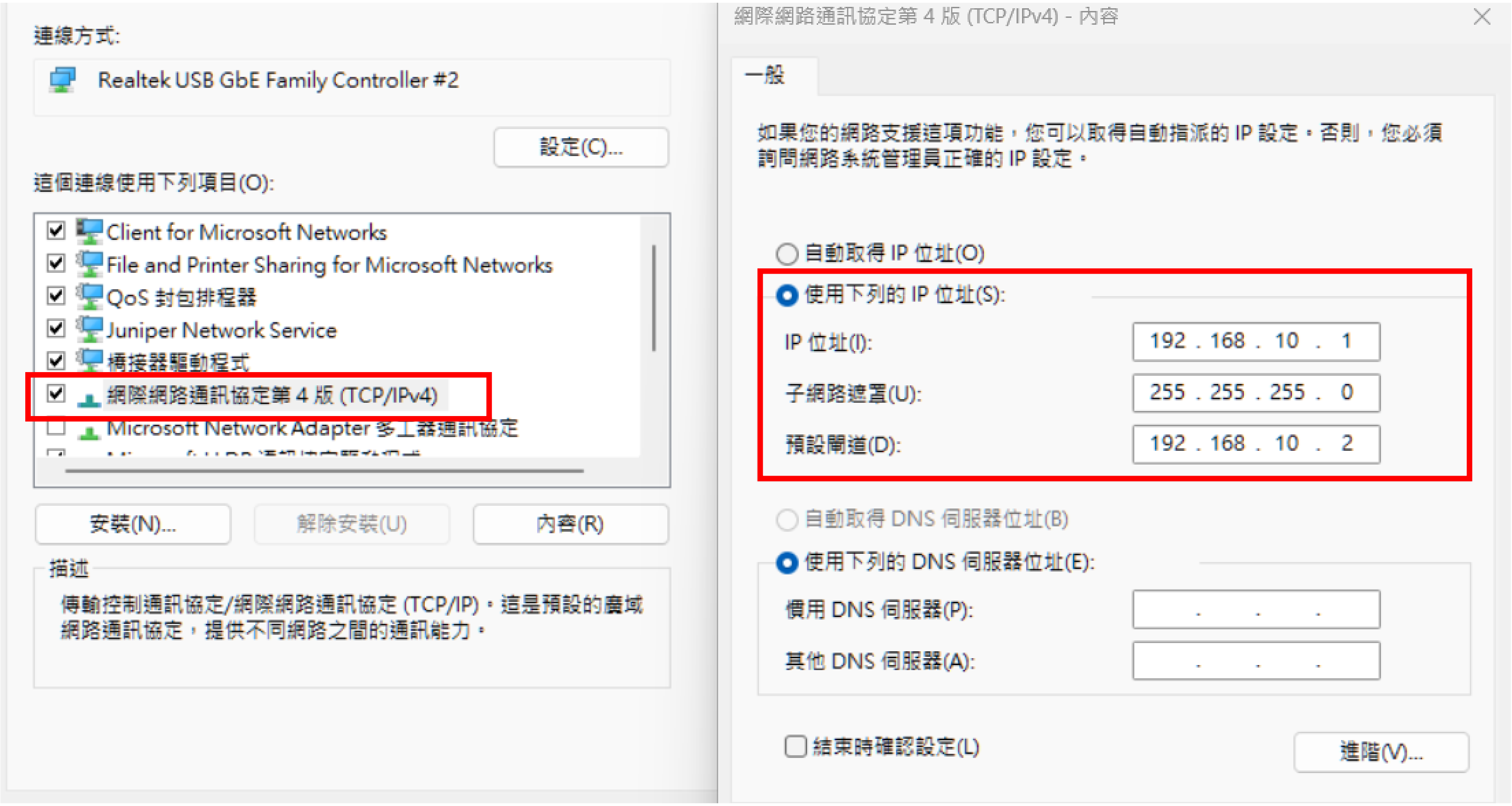
USRP Preparation - Hardware

- One USRP-2920 machine
- One power cable
- Two Antennas
- One gigabyte ethernet cable
- USB to ethernet adapter (if needed)

USRP Preparation – Connection Setting

- We need to configure the internet setting between USRP and computer manually

USRP Preparation – Connection Setting



Hello World in USRP

- Using the command ***findsdru()***
- You should get the following result

```
ans =

  struct with fields:

     Platform: 'N200/N210/USRP2'
    IPAddress: '192.168.10.2'
    SerialNum: '4095'
       Status: 'Success'
```

Reference

Transmit & Receive Scheme

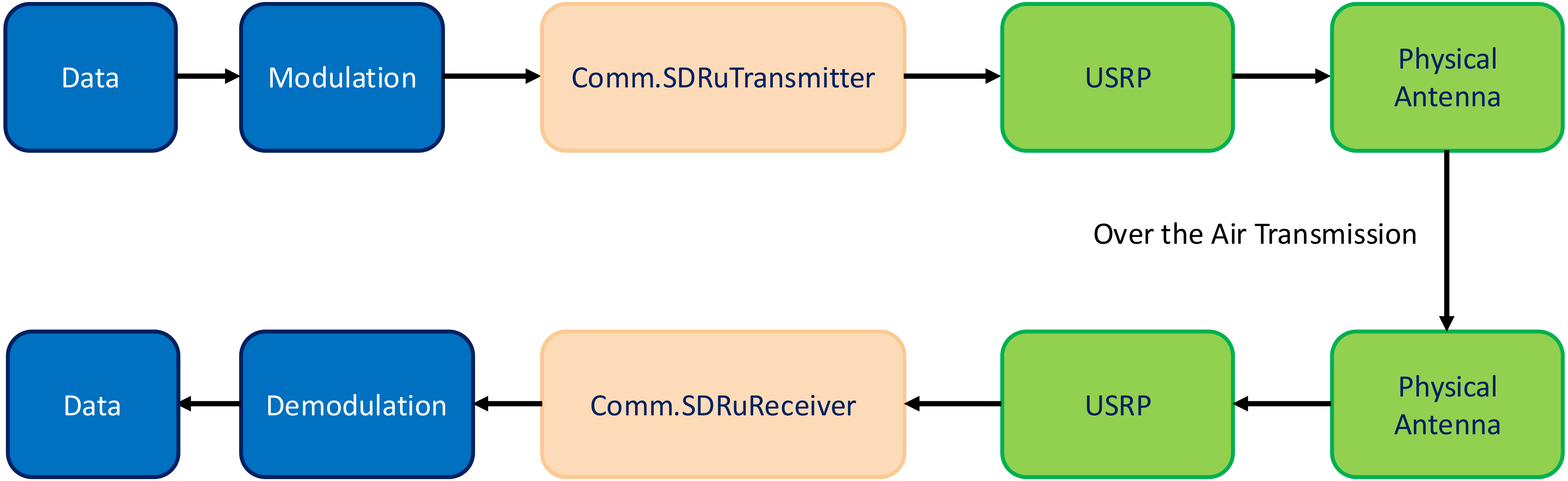
- In Matlab, we communicate with USRP using comm.SDRuTransmitter and comm.SDRuReceiver.

```
radio_Tx = comm.SDRuTransmitter(...
                'Platform',              platform, ...
                'IPAddress',             address, ...
                'CenterFrequency',       USRPCenterFrequency, ...
                'Gain',                  USRPGain);
```

Transmit & Receive Scheme

```
radio_Rx = comm.SDRuReceiver(...
                'Platform',             platform, ...
                'IPAddress',            address, ...
                'CenterFrequency',      USRPCenterFrequency, ...
                'Gain',                 5, ...
                'SamplesPerFrame',      1e5, ...
                'OutputDataType',       'double') ;
```

USRP Transmission Scheme

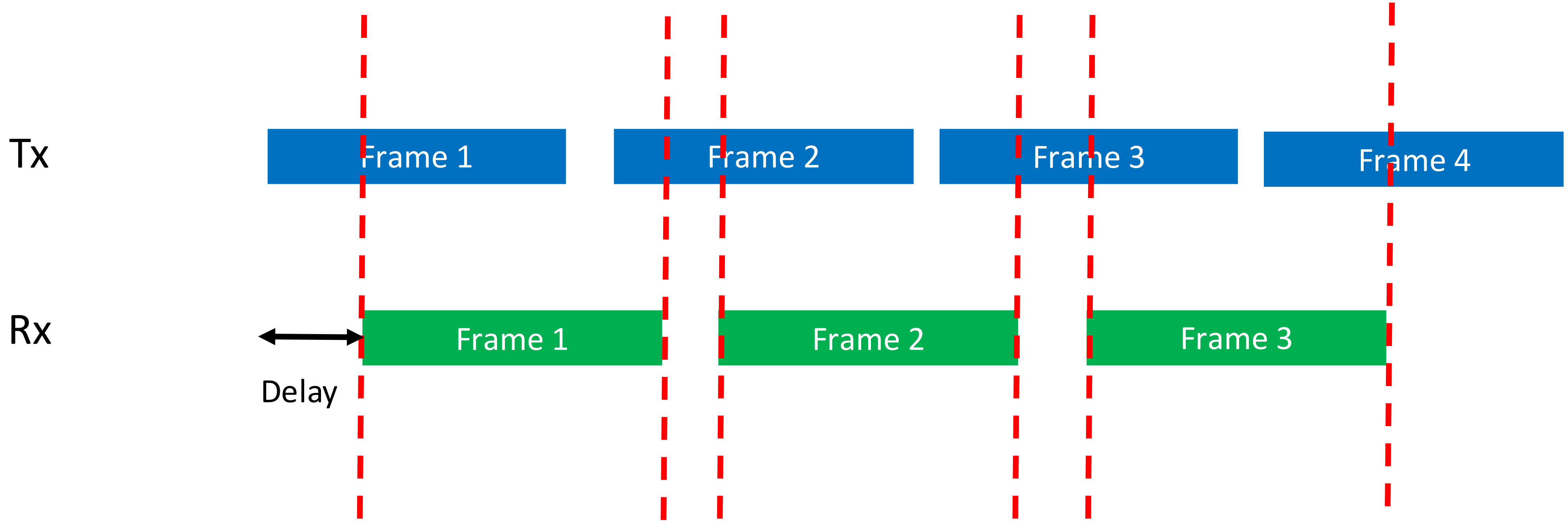


Basic Example - Transmitting a sine wave

- Download the sample code from NTU cool
- You should be aware of
  - Timing of transmission and reception
  - Transmission Delay of USRP

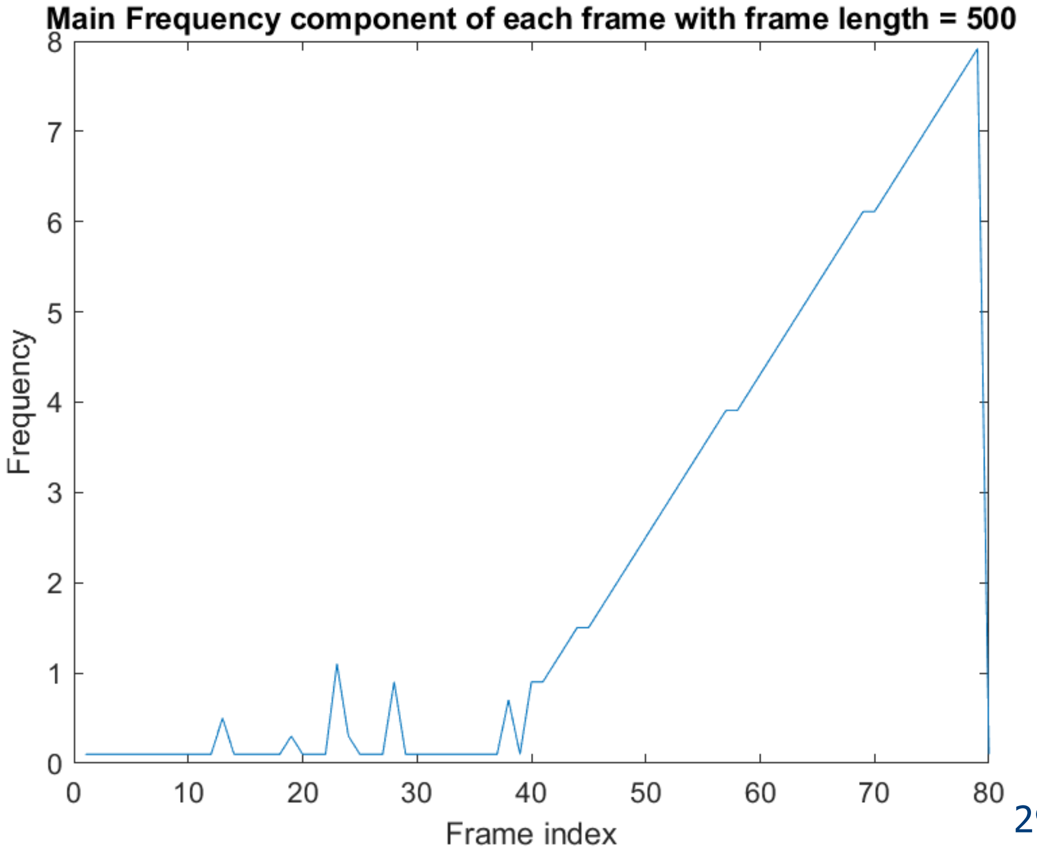
USRP Transmission – Timing

- Timing issue of Tx and Rx



Transient state of USRP

- In the sample code, we transmit sine wave with different frequency for 50 frames
- $f_i = \frac{Frame\ index}{10}$
- We keep reception for 30 frames



Center Frequency

Group 1	Group 2	Group 3	Group 4
875MHz	885MHz	895MHz	905MHz
Group 5	Group 6	Group 7	Group 8
915MHz	925MHz	935MHz	945MHz

VERT900 Antenna

$93.00 USD

782773-01 QTY: 1 Add to Part List

VERT900 Vertical Antenna (824-960 MHz, 1710-1990 MHz) Dualband

Includes one VERT900 824 to 960 MHz, 1710 to 1990 MHz Quad-band Cellular/PCS and ISM Band omni-directional vertical antenna, at 3dBi Gain.

Master Clock rate

Master clock rate in Hz, specified as a positive scalar. The master clock rate is the analog to digital (A/D) and digital to analog (D/A) clock rate. The valid range of values for this property depends on the connected radio platform.

`Platform` Property Value	`MasterClockRate` Property Value (in Hz)
`"N200/N210/USRP2"`	`100e6` (read-only).
`"B200"` or `"B210"`	Scalar in the range from `5e6` to `61.44e6`. When you use a B210 radio with multiple channels, the clock rate must be less than or equal to 30.72e6. This restriction is a hardware limitation for two-channel operations on B210 radios. The default value is `32e6`.

Decimation / Interpolation

Default: 512